

SAFETY DATA SHEET

BioPura™ renewable MEG

Date: 20.09.2023

Version: 1.0

Former date: ----

Former version: ----

SECTION 1: IDENTIFICATION OF SUBSTANCE AND OF THE COMPANY/UNDERTAKING

1.1 Product identifier

Product name: BioPura™ renewable MEG
Substance name: Ethylene glycol
EC No: 203-473-3
CAS No: 107-21-1
INDEX No: 603-027-00-1
UK REACH registration number: UK-01-1060922537-9-xxxx

1.2 Relevant identified uses of the substance and uses advised against

Identified uses: Laboratory chemicals, production of substances
Uses advised against: -

1.3 Details of the supplier of the safety data sheet

Information of the manufacturer: UPM-Kymmene Oyj
UPM Biochemicals Sales GmbH
FI-53200 Lappeenranta, Finland
glycols@upm.com
+49 3461 519 5000

1.4 Emergency telephone number: Emergency call UPM Biochemicals GmbH
+49 3461 519 5001, daily (Monday to Friday) from 9:00 a.m. to 5:00 p.m. available

National Poisons Information Service (NPIS)
NHS 111 (England), NHS 24 (Scotland) or NHS Direct (Wales) – dial 111
In Northern Ireland contact your local GP
Healthcare Professionals: UK NPIS 0344 892 0111

SECTION 2: HAZARDS IDENTIFICATION

2.1 Classification of the substance

Classification according to Regulation (EC) No 1272/2008 (CLP):
Acute toxicity, Oral (Category 4), H302
Specific target organ toxicity — repeated exposure, Oral (Category 2), Kidneys, H373
Please refer to section 16 for the full text of the hazard statements listed in this section.

2.2 Label elements

Classification in accordance with Regulation (EC) No 1272/2008

Pictogram



Signal word:

Warning

Hazard designation(s):

H302
H373

Harmful if swallowed.
May cause damage to organs (kidneys) through prolonged or repeated exposure if swallowed.

Precautions:

P301 + P312 + P330

IF SWALLOWED: Call a POISON CENTRE/doctor if you feel unwell. Rinse mouth.

2.3 Additional hazards

This substance/mixture does not contain any components at concentrations of 0.1% or higher that are classified either as persistent, bioaccumulative and toxic (PBT) or as very persistent and very bioaccumulative (vPvB).

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SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

3.1 Substance

Formula: C₂H₆O₂

Molecular weight: 62.07 g/mol

CAS No: 107-21-1

EC No: 203-473-3

INDEX No: 603-027-00-1

UK REACH registration number: UK-01-1060922537-9-xxxx

EC no	CAS no	CA Index Name	Concentration	Classification according to Regulation (EC) No 1272/2008
203-473-3	107-21-1	Ethane-1,2-diol	≥ 99.9%	Acute Tox. 4 H302 ^(*) / ^{**} STOT RE 2; H373

^(*) Harmonised classification and labelling

^{**} Classification specified by the manufacturer that includes other classification in addition to the classification specified by Regulation (EC) No 1272/2008

Please refer to section 16 for the full text of the hazard statements listed in this section.

SECTION 4: FIRST AID MEASURES

4.1 Description of first aid measures

General directions

Seek medical attention. Show the attending physician this safety data sheet.

After inhalation

If inhaled, move the affected person to the fresh air. Begin artificial respiration if breathing has stopped. Seek medical attention.

After skin contact

Wash off with soap and plenty of water. Seek medical attention.

After eye contact

Carefully rinse eyes with water.

After swallowing

Never pour anything into the mouth of an unconscious person. Wash out mouth with water. Seek medical attention.

4.2 Most important symptoms and effects, both acute and delayed

The most known symptoms and side effects are described on the identification label (see section 2.2) and/or in section 11.

4.3 Indication of any immediate medical attention and special treatment needed

No data available.

SECTION 5: FIREFIGHTING MEASURES

5.1 Extinguishing media

Suitable extinguishing media: Adapt fire-fighting measures to the environment.
Use sprayed water, alcohol-resistant foam, dry chemical or carbon dioxide.

5.2 Special hazards arising from the substance or mixture

No data available.

5.3 Advice for firefighters

In case of a fire, if necessary, wear flame-proof clothing and self-contained breathing apparatus.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal precautions, protective equipment and emergency procedures

Allow only well-trained experts wearing suitable protective clothing to abide in the field of the accident. Do not breathe in vapours/mist/gas. Ensure there is sufficient ventilation. Remove persons to safety.

See section 8 for information on personal protective equipment.

6.2 Environmental precautions

Dispose of the spillage and the resulting waste according to the applicable environmental regulations. Do not allow the product and the resulting waste to enter sewers/soil/surface or ground water. Notify the respective authorities in accordance with local law in the case of environmental pollution immediately.

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6.3 Methods and materials for containment and cleaning up

Soak up with inert absorbent and dispose of as waste requiring monitoring. Keep in suitable, closed containers for disposal.

6.4 Reference to other sections

For further and detailed information see section 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for safe handling

Avoid contact with skin and eyes. Avoid inhaling the vapours or mist.

Please refer to section 2.2 for information on protective measures.

7.2 Conditions for safe storage, including any incompatibilities

Store in a cool place. Store in a dry, well-ventilated place in tightly closed containers. Carefully close opened containers and store upright to prevent leakage.

Hygroscopic.

Storage class (TRGS 510): Flammable liquids

7.3 Specific end use(s)

No specific applications other than those listed in section 1.2 are planned.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

8.1 Control parameters

Occupational exposure limit values (EH40/2005 Workplace Exposure Limits):

Long Term Exposure Limit (8-hr TWA reference period): 20 ppm, 52 mg/m³; Short-term Exposure (15-minutes reference period): 40 ppm; 104 mg/m³

Components with workplace-related, monitored limit

Ingredients	CAS No	Value/type of the exposure	Control parameters	Basis
Ethylene glycol	107-21-1	STEL	40 ppm 104 mg/m ³	Commission Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
	Comments	Shows the chance of large amounts of the substance being absorbed through the skin Indicative		
		TWA	20 ppm 52 mg/m ³	Commission Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values
		Shows the chance of large amounts of the substance being absorbed through the skin Indicative		
		OEL (Vapours and aerosols)	10 ppm 26 mg/m ³	TRGS 900 - Occupational exposure limits
		Senate Commission for the Investigation of Health Hazards of Chemical Compounds in the Work Area of the German Research Foundation (DFG) (MAK commission) European Union (The EU has defined an air threshold: deviations in the value and peak limit are possible.) Sum of vapours and aerosols. Absorbed through skin There is no need to fear a risk of foetal impairment if the occupational exposure limit value and the biological limit value (BGW) are observed		

Derived no-effect level (DNEL)

Scope	Exposure routes	Effects on health	Value
Employee	Inhalation	Long-term – local effects	35 mg/m ³
Employee	Skin contact	Long-term – systemic effects	106 mg/kg body weight/day
Consumer	Inhalation	Long-term – local effects	7 mg/m ³
Consumer	Skin contact	Long-term – systemic effects	53 mg/kg body weight/day

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Predicted no-effect concentration (PNEC)

Compartment	Value
Soil	1.53 mg/kg
Sea water	1 mg/l
Fresh water	10 mg/l
Marine sediment	3.7 mg/kg
Fresh water sediment	37 mg/kg
Waste water processing plant	199.5 mg/l
Sporadic release in water	10 mg/l

8.2 Exposure controls

In case of a hazardous material with no controlled concentration limit it is the employer's duty to keep concentration levels down to a minimum achievable by existing scientific and technological means, where the hazardous substance poses no harm to workers.

Appropriate engineering controls

The usual precautionary measures when handling chemicals must be observed. Wash hands before breaks and at the end of work.

Individual protection measures, such as personal protective equipment

The general hygiene measures at the workplace must be observed.

Observing the principles of safety and hygiene in the workplace, personal protective equipment (PPE) should be used in combination with other control measures.

Observe the usual precautionary measures when handling the product.

Wearing closed work clothes is recommended.

Keep away from food, drink and animal feed.

Immediately remove soiled, saturated clothing.

Wash hands before breaks and at the end of work.

Avoid contact with eyes and skin.

Eyeface protection

Facial protection and safety glasses. For eye protection, use only equipment that has been tested and approved according to regulatory standards such as NIOSH (US) or EN 166 (EU).

Skin protection

Hand protection

Work with gloves. Check gloves before use. Use an appropriate method to remove gloves (without touching the outer glove surface) to avoid skin contact with this product. Disposal of contaminated gloves after use within the scope of legal requirements and good laboratory practice. Wash and dry hands.

The protective gloves to be used must meet the specifications of EC Directive 89/686/EEC and the resultant standard EN 374.

Full contact

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Breakthrough time: 480 min

Material tested: Dermatrill® (KCL 740 / Aldrich Z677272, size M)

Spray contact

Material: Nitrile rubber

Minimum layer thickness: 0.11 mm

Breakthrough time: 480 min

Material tested: Dermatrill® (KCL 740 / Aldrich Z677272, size M)

Data source: KCL GmbH, D-36124 Eichenzell, Telephone +49 (0)6659 87300, email: sales@kcl.de,

Test method: EN374

You must contact the supplier of CE-approved gloves if dissolving in or mixing with other substances and if conditions differ from EN 374. This recommendation is considered advice and must be assessed by an industrial hygienist and a safety professional who know the specific situation of the intended use by our customers. It should not be construed as signifying approval for each specific use.

Other

Select body protection equipment depending on the activity and possible exposure. Wear general protective clothing to minimise skin exposure and contamination. Do not take work clothes home and wash them daily. Apply water-insoluble skin protection preparations to clean hands and rub in thoroughly before starting work and after each break. Before breaks and at the end of work, skin must be cleaned with soap and water; after cleaning, use skin care products containing grease.

Respiratory protection

If according to the risk assessment, air-purifying respiratory protection is required, a full-face mask with multi-purpose combination filter (US) or with filter type ABEK (EN 14387) must be used in addition to the technical measures. If the respiratory equipment is the only protective measure, use self-contained breathing apparatus with a full-face mask. Respiratory equipment and components must be tested and approved according to appropriate government standards such as NIOSH (US) or CEN (EU).

Thermal hazards

No thermal hazards known.

Environmental exposure controls

If safe to do so, avoid further leakage or spillage. Do not empty into drains.

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The requirements detailed in section 8 assume skilled work under normal conditions and usage of the product for appropriate aims. If conditions differ from normal or work is carried out under extreme conditions, an expert's advice is necessary before deciding upon further protective measures.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

9.1 Information on basic physical and chemical properties

Appearance:	Form: liquid Colour: colourless
Odour:	No data available.
Odour threshold:	No data available.
pH:	No data available.
Melting point/freezing point:	-12.99 °C
Initial boiling point and boiling range:	195 - 198 °C at 1,013 hPa
Flash point:	111 °C - closed crucible
Evaporation rate:	1
Flammability (solid, gas):	No data available.
Upper/lower flammability or explosive limits:	Upper explosive limit: 15.3%(V) Lower explosive limit: 3.2%(V)
Vapour pressure:	0.11 hPa at 20 °C
Vapour density:	2.14 - (Air = 1.0)
Relative density:	1.1130 g/cm ³
Water solubility:	Fully miscible soluble.
Partition coefficient: n-octanol/water:	log Pow: -1.36
Auto-ignition temperature:	Auto-flammability at 400 °C
Decomposition temperature:	No data available.
Viscosity:	No data available.
Explosive properties:	No data available.
Oxidising properties:	No data available.

9.2 Other information

Relative vapour density 2.14 - (Air = 1.0)

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

No data available.

10.2 Chemical stability

Stable under the defined storage conditions.

10.3 Possibility of hazardous reactions

No data available.

10.4 Conditions to avoid

No data available.

10.5 Incompatible materials

Strong acids, strong oxidizing agents, strong bases, aldehydes, aluminium.

10.6 Incompatible materials

Hazardous decomposition products may develop in case of fire. - Carbon oxides

Other decomposition products - No data available.

In the event of fire: See section 5.

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SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on toxicological effects

Acute toxicity

LD₅₀ oral - Rats - 4,700 mg/kg

LD₅₀ skin - Rabbit - 10,626 mg/kg

Skin corrosion/irritation

Skin – Rabbit

Result: No skin irritation.

Serious eye damage/irritation

Eyes – Rabbit

Result: Slight eye irritation - 24h

Respiratory or skin sensitization

No data available.

Germ cell mutagenicity

No data available

Carcinogenicity

According to IARC, ACGIH, NTP or EPA classifications, this product is or contains an ingredient classified as a probable non-carcinogen.

IARC: None of the components of this product present at a concentration equal to or greater than 0.1% have been identified by IARC as a probable, possible or proven human carcinogen.

Reproductive toxicity

Teratogenic effects have been demonstrated in laboratory tests.

According to animal studies, exposure above the limit value may lead to reproductive disorders.

Specific target organ toxicity - single exposure

No data available.

Specific target organ toxicity - repeated exposure

Oral - May cause damage to organs through prolonged or repeated exposure. – Kidneys

Aspiration hazard

No data available.

Additional Information

RTECS: KW2975000

On ingestion, early symptoms resemble alcoholic intoxication and are followed by nausea, vomiting, abdominal pain, weakness, muscle sensitivity, respiratory failure, convulsions, cardiovascular collapse, pulmonary oedema, hypocalcemia-induced tetany and severe metabolic acidosis. Without treatment, death can occur in 8 to 24 hours. Victims who survive the initial poisoning phase tend to develop kidney failure, along with brain and liver damage., Exposure to and/or ingestion of alcohol may potentiate the toxic effects.

Central nervous system - Irregularities - Based on evidence in humans.

SECTION 12: ECOLOGICAL INFORMATION

12.1 Toxicity

Toxicity to
fish:

LC₅₀ - Oncorhynchus mykiss (rainbow trout) - 18,500 mg/l - 96h

LC₅₀ - Leuciscus idus (golden orfe) - > 10,000 mg/l - 48h

NOEC - Pimephales promelas (fathead minnow) - 32,000 mg/l - 7 d

NOEC - Pimephales promelas (fathead minnow) - 39,140 mg/l - 96h

Toxicity to

Daphnia and other aquatic invertebrates:

EC₅₀ - Daphnia magna (large water flea) - 74,000 mg/l - 24h

NOEC - Daphnia (water flea) - 24,000 mg/l - 48h

LC₅₀ - Daphnia magna (large water flea) - 41,000 mg/l - 48h

12.2 Persistence and degradability

No data available.

Ratio of BOD/ThBOD 0.78%.

12.3 Bioaccumulative potential

Bioaccumulation is not expected based on the known physicochemical properties of the substance.

No bioaccumulation.

Bioaccumulation in other fish - 61 d

- 50 mg/l

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12.4 Mobility in soil

No data available.

12.5 Results of PBT and vPvB assessment

No data available.

12.6 Endocrine disrupting properties

This substance/mixture does not contain any components at concentrations of 0.1% or higher that are classified either as persistent, bioaccumulative and toxic (PBT) or as very persistent and very bioaccumulative (vPvB).

12.7 Other adverse effects

No data available.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste treatment methods

Disposal according to the local regulations.

Information regarding the disposal of the product

Dispose of in accordance with applicable regulations.

Use a recognized disposal company to dispose of residues and non-recyclable solutions.

Information regarding the disposal of the packaging

Dispose of as used product. Dispose of in accordance with applicable regulations.

SECTION 14: TRANSPORT INFORMATION

14.1 UN number or ID number

Not applicable.

14.2 UN proper shipping name

No hazardous material.

14.3 Transport hazard class(es)

Not applicable.

14.4 Packing group

Not applicable.

14.5 Environmental hazards

Environmentally hazardous: No.

Marine pollutant: No.

14.6 Special precautions for user

See section 7.

14.7 Maritime transport in bulk according to IMO instruments

Not applicable.

Section 15: REGULATORY INFORMATION

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

This Safety Data Sheet complies with the requirements of Regulation (EC) No 1907/2006.

REGULATION (EC) No 1907/2006 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH), establishing a European Chemicals Agency, amending Directive (EC) No 1999/45 and repealing Council Regulation (EEC) No 793/93 and Commission Regulation (EC) No 1488/94 as well as Council Directive (EEC) No 76/769 and Commission Directives (EEC) No 91/155, (EEC) No 93/67, (EC) No 93/105 and (EC) No 2000/21

REGULATION (EC) No 1272/2008 OF THE EUROPEAN PARLIAMENT AND OF THE COUNCIL of 16 December 2008 on classification, labelling and packaging of substances and mixtures, amending and repealing Directives (EEC) No 67/548 and (EC) No 1999/45, and amending Regulation (EC) No 1907/2006

COMMISSION REGULATION (EU) 2020/878 of 18 June 2020 amending Annex II to Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH)

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Regulation (EU) No 649/2012 concerning the export and import of hazardous chemicals (PIC):
Not listed.
Regulation (EC) 1005/2009 on substances that deplete the ozone layer:
Not listed.
Regulation (EC) 648/2004 on detergents (Detergents Regulation):
Not applicable.
Regulation (EU) 2019/1021 on persistent organic pollutants (POP):
Not listed.
Restrictions according to Annex VIII of Regulation (EC) No 1907/2006:
Not listed.

National regulations

Water hazard class:

WGK 1 (German water hazard class), slightly hazardous to water - identification number 105 - VwVwS, (Administrative Regulation on the Classification of Substances Hazardous to Waters), The assessment refers to the material without any additives. If additives are added, higher WGKs are possible according to the rules specified in Annex 4 (classification of mixtures into water hazard classes).

15.2 Chemical safety assessment

A chemical safety assessment has been carried out for this substance.

SECTION 16: OTHER INFORMATION

Information regarding the revision of the safety data sheet

Previous Version:	1.0/EN	Previous date:	05.07.2023
Version:	1.0/GBR	Date:	20.09.2023
Nature of the change:	New version.		
Reason:	Amendment to REACH and CLP Regulation.		

Relevant hazard statements (code and full text) of sections 2 and 3

H302 - Harmful if swallowed.

H373 - May cause damage to organs (kidneys) through prolonged or repeated exposure if swallowed.

Training advice

Follow the safety data sheet when handling this substance.

Full test of the abbreviation in the safety data sheet

ACGIH: American Conference of Governmental Industrial Hygienists.

ADN: European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways.

ADR: Agreement concerning the International Carriage of Dangerous Goods by Road.

BOD: Biochemical Oxygen Demand.

CAS number: Chemical Abstract Service number.

CEN: European Committee for Standardization.

CLP: Regulation (EC) No 1272/2008 on classification, labelling and packaging of substances and mixtures.

EC: European Community.

EC number: EINECS and ELINCS numbers (see also EINECS and ELINCS).

EINECS: European Inventory of Existing Commercial Chemical Substances.

ELINCS: European List of Notified Chemical Substances.

EN: European Norm.

EPA: United States Environmental Protection Agency.

EU: European Union.

EWC: European Waste Catalogue (replaced by LoW – see below).

GHS: Globally Harmonized System of Classification and Labelling of Chemicals.

IARC: International Agency For Research On Cancer.

IATA: International Air Transport Association.

ICAO-TI: Technical Instructions for the Safe Transport of Dangerous Goods by Air.

IMDG: International Maritime Dangerous Goods.

IMO: International Maritime Organization.

IMSBC: International Maritime Solid Bulk Cargoes.

Kow: n-Octanol - Water Partition Coefficient.

LC₅₀: Lethal concentration resulting in 50 % mortality.

LGK: Storage class (Germany).

LoW: List of Waste.

MAK: Threshold limit value.

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NIOSH: National Institute for Occupational Safety & Health.
NOEL: No observed effect level.
NTP: National Toxicology Program (NTP).
OECD: Organization for Economic Cooperation and Development.
OEL: Occupational exposure limits.
PBT: Persistent, Bioaccumulative and Toxic.
PPE: Personal Protective Equipment.
REACH: Regulation 1907/2006/EC concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals.
SDS: Safety Data Sheet.
STEL: Short-Term Exposure Limit.
STOT: Specific Target Organ Toxicity.
TA: Technical guidance (Germany).
ThBOD: Theoretical Biochemical Oxygen Demand.
TRGS: Technical rules for hazardous substances (Germany).
TWA: Time-Weighted Average.
UN: United Nations.
WGK: Water hazard class (Germany).
vPvB: very Persistent and very Bioaccumulative.

This safety data sheet had been prepared on the basis of information provided by the manufacturer/supplier and conform to the relevant regulations. The information, data and recommendations contained herein are provided in good faith, obtained from reliable sources and believed to be true and accurate as of the date issued; however, no representation is made as to the comprehensiveness of the information. The SDS shall be used only as a guide for handling the product; in the course of handling and using the product other considerations may arise or be required. Users are cautioned to determine the appropriateness and applicability of the above information to their particular circumstances and purposes and assume all risk associated with the use of this product. It is the responsibility of the user to fully comply with local, national and international regulations concerning the use of this product.